



High Directional & Ultra-Wideband Antenna

MAJOR PROJECT SYNOPSIS REPORT

**Submitted in Partial Fulfillment of the Requirements of the Degree of Bachelor of
Technology in Electronics and Telecommunication Engineering**

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CERTIFICATE

*This is to certify that the project entitled “**High Directional & Ultra-Wideband Antenna**” is a bonafide work of students Dhandge Jayesh, Dhekale Yash, Borse Mayur submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Electronics and Telecommunication Engineering.*

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For
Bachelor of Technology in Electronics and
Telecommunication Engineering

This Project report entitled “**High Directional & Ultra-Wideband Antenna**”

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ACKNOWLEDGEMENT

A successful work of dissertation is the result of inspiration, support, guidance, motivation and co-operation of facilities during study. It gives a great pleasure to acknowledge our deep sense of gratitude to present our project titled; “High Directional & Ultra-Wideband Antenna”. We would like to give sincere thanks to our Principal Dr. Vivek Sunnapawar and H.O.D. Dr. Jayashree Khanapuri for giving me opportunity to present this seminar. I am also thankful to my respected guides Prof. Martand Jha & Prof. Kiran Rathod for their whole hearted support and affectionate encouragement. Their dynamism, vision, sincerity have deeply inspired us. They have taught us the methodology to present the research work as clearly as possible.

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ABSTRACT

There is a significant demand for high-performance, efficient antennas in various applications such as telecommunications, defense, and advanced wireless networks where fast data transmission and accurate control of signal is required. This design project consists of a high directional, ultra-wide band antenna at a frequency of 5-6 GHz. The design is targeting a gain of 12-15 dBi, bandwidth of 1 GHz, and a narrow beamwidth of 30° in both azimuth and elevation with up to 100W of power handling capacity, hence robust performance for high-power applications.

The 5-6 GHz band is widely utilized in Wi-Fi, radar, and communication systems because of its balance between range and data capacity. It is well suited for applications requiring stable, high-speed connections with minimal interference. To meet these demands, the antenna is designed with a linear polarization to ensure signal clarity and reduce cross-polarization interference. This research spans a wide bandwidth to show the high-gain and consistent beamwidth challenges in maintaining a practical solution for next-generation communication systems.

This report contains information about the design methodology, working principles, expected performances, and probable applications concerning the antenna designed, hence of interest in both business and specialized purposes. Designed and tested, this was one of the high-directive and wideband high-frequency antennas created, placing the basis for future works concerning antenna research.

INTRODUCTION

Lately, there has been strong thrust for the development of high-performance antennas that should effectively operate in high frequencies, particularly in the range of GHz. Antennas have a crucial role because of their ability to pass along and receive electromagnetic waves with distances covered over a mile or even more, serving industries as wide-ranging as telecommunication and defense industries. With the advancement in technology and increased demand for fast data transfer, the need for antennas with high gain and wide bandwidth in certain frequency bands has also increased. This project aims to design a high-directional, wideband antenna working in the 5-6 GHz frequency band and efficiently meet the industrial requirements.

- **Importance of 5-6 GHz Frequency Band**

The 5-6 GHz frequency band is gaining usage due to its balance of range and data handling capacity. This band is increasingly used in Wi-Fi communications, satellite communications, radar systems, and advanced wireless sensor networks. Specifically, wi-fi networks make use of portions of the 5 GHz band due to limited interference, since this band is typically not very crowded compared to the 2.4 GHz band. The systems operating at this frequency band will provide more data rates and lower latencies, which are considered critical for applications that have the need to exchange large amounts of data rapidly. Moreover, the radar system also benefits by achieving good range resolution and target detection capabilities, which are extremely important in both civil and military applications.

- **Challenges in Designing High-Directional and Wideband Antennas**

Engineeringly, the design of a highly directive antenna which radiates with high directionality over a wide frequency range is an extremely tough challenge. It is very typical that, in order to achieve such a gain of 12-15 dBi, the energy should be concentrated into one particular direction only, thereby indicating a much narrower beamwidth. While in this case, the critical narrow beamwidth at 30° in both azimuth and elevation must be maintained over a wide bandwidth of 1 GHz, by careful optimization of the structure and materials of the antenna as well as design parameters. This is the kind of accuracy required to ensure that the antenna will have major losses neither in gain nor loss of signal integrity over the operating frequency range.

OBJECTIVE

This paper aims at the design and development of a high-directional ultra-wideband antenna operating in the 5-6 GHz frequency band, targeting the increasing trend of using high-performance antennas in modern communication, defense, and wireless network systems. Given the fact that most industries are dependent on the transmission of data from one end to another without any interruptions, at very high speed, this paper is trying to meet up with the following design requirements that should make the optimized performance, efficiency, and adaptability of the designed antenna possible:

1. High Directional Gain (12-15 dBi)

The first objective is to design an antenna with high directional gain between 12-15 dBi. This high gain will make the antenna concentrate its radiated energy in a particular direction, thus making it capable of transmitting and receiving signals over longer distances with stronger signal strength.

High directional gain is required in applications demanding long range and focused transmission of signal. These are radar systems, point to point communication, and applications of Wi-Fi where the interference should be minimized.

2. Wideband Performance (1 GHz Bandwidth)

The antenna needs to be designed to work effectively in a 1 GHz bandwidth and the frequency range 5-6 GHz. It needs to be wideband enough to accommodate high data rates and several frequency channels within this range. It is very much applicable to modern communication systems, which require robust data transmission.

It is difficult to design as it must maintain stable gain, impedance matching, and minimal return loss across the entire bandwidth, thus demanding very careful engineering in order to maintain consistent performance across a very wide frequency range.

3. Optimized Beamwidth of 30° in both Azimuth and Elevation

Another objective is the half-power beamwidth that occurs at both azimuth and elevation planes and is supposed to be 30°. This means that it will tightly focus the antenna's radiation pattern. Narrow beamwidth is useful for the antenna as it eliminates interference from any source, thus increasing efficiency as the signal is concentrated in a defined area.

A narrow beam width is very useful in applications demanding precise, targeted communication, such as radar and point-to-point wireless communication, where the accuracy of the signal is critical.

4. To 100W Power Handling Capability

Must be able to handle an input power of up to 100W without degrading performance or reliability. More power handling is required any time the application requires a signal that is more powerful in order to have a meaningful signal or if it had to operate in an environment in which there was significant loss of signal.

Effective handling of power should be applied in the design of an antenna to deal with its thermal loads in order to avoid overheating, something that could degrade performance otherwise or damage its components progressively.

5. Linear Polarization for Signal Integrity

The linear polarization design is utilized in the antenna to avoid interference caused by cross-polarization and to ensure that the signal integrity is maintained upon transmission or reception. The advantage of linear polarization in many communication systems is the minimization of unwanted polarization mismatches, which can lead to increased signal clarity.

In choosing a linear polarization, this design maximizes compatibility with systems already utilizing similar polarization characteristics, making the antenna's practical applications more significant.

6. Compatibility with Sophisticated Communication and Sensor Systems

The Antenna should be compatible with several high-tech applications such as Wi-Fi, Radar and even wireless sensor networks. Such adaptability will allow it to be useful in numerous industrial applications, within both the private and specialized realms.

Designs for the future integration into emerging technologies like 5G and 6G networks, IoT based sensor systems, and also radar technologies so that these antennas would be able to integrate with any new wireless communications and data transfer.

7. Robust and Efficient Design

The basic objective is to develop such a configuration that will have the maximum performance with physical strength. The antenna should not degrade easily by environmental exposure, for example, temperature fluctuations and mechanical loading, without losing any operating efficiency.

To ensure reliable long-term deployment in a variety of environments, material selection, structural strength, and geometrical configuration are considered so that the performance is retained over time.

LITERATURE SURVEY

Research Paper Name, Author	Observations	Key Findings
A Novel HF Fishbone antenna with VEE element Author: -Zhigang Zhang, Hui Xie, Longgen Zheng, Qin Wang	In this research paper the novel HF fishbone antenna with V-shaped dipoles offers a 2-5dB higher directivity at higher frequencies compared to traditional fishbone antennas. It maintains low side-lobe levels and a wide operating band. The optimized parameters for arm length, open angle, and spacing significantly enhance performance.	Longer dipole arms present structural challenges, and the directivity improvement plateaus beyond certain lengths, indicating limitations in achieving further gains.
Simulation and analysis of arrays formed by HF fishbone antenna Author: -Zhigang Zhang, Li Yin & Ting Li	In the research we found that two-element fishbone sector arrays are optimal for low-frequency beam scanning. Four-element and six-element arrays, particularly in parallel and sector configurations, suffer from increased sidelobes and reduced directivity gain at higher phase shifts and frequencies.	A major drawback is the narrow main lobe in high-frequency ranges, causing ineffective array performance, particularly for large arrays like 1x6 parallel configurations. Additionally, phase shifts at high frequencies can lead to lobe division and multiple sidelobes.
Design and Simulation of Ultra-Wideband Antennas for Wireless Applications	In the research we found that two-element fishbone sector arrays are optimal for low-frequency beam scanning. Four-element and six-element arrays, particularly in parallel and	This study explores the development of UWB antennas with stable gain and minimal interference over the 5-6 GHz band, crucial for short-range wireless communication. It

Author: -Madhumitha Jayaram; Sowmiya Murugan; Shamilee Mohandass; Bhagyaveni Marchala Anjaneyulu	sector configurations, suffer from increased sidelobes and reduced directivity gain at higher phase shifts and frequencies.	focuses on optimizing radiation patterns to improve data transmission rates while minimizing design complexity. The proposed designs use simulation software for performance validation, yielding highly efficient antennas for high-speed wireless applications.
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SOCIAL/ MEDICAL/ AGRICULTURAL/ INDUSTRY RELEVANCE

A wideband antenna that is directional and high with the specifications suited to the frequency range of 5-6 GHz has far reaching applications across many sectors. Advancement in communication technology requires efficient, high-performance antennas as a means to address the needs of the modern society through healthcare innovation to increased industrial automation. Below is an extension of the social, medical, agricultural, and industrial relevance of this antenna that will portray ways in which such technology would spur on progress, increase efficiency, and solve critical problems.

1. Social Relevance: Improving Accessibility and Connectivity

In the social field, reliable and efficient antennas are required since connectivity has to be taken to those underserved remote places. High- directional antennas operating in a 5-6-GHz frequency band can make high speed, long- range wireless communication possible thus enhancing Internet and network accesses. For such an application with high gain coupled with low beamwidth values, applications include:

- **Rural Broadband Extension:** With support for point-to-point wireless communication, it can further extend broadband availability to the rural or distant areas that are otherwise hard to provide with a fixed wired infrastructure.
- **Public Wi-Fi Network:** For urban use cases, installation of high gain antennas offers strong and broad coverage of public space where the public demand free accessible internet access in public spaces like parks, libraries, and community centres.
- **Emergency Communication Systems:** In natural disasters and other emergencies, the urgency to communicate promptly and efficiently determines relief operation organization. With this antenna's directional characteristic, there will be stable connectivity between mobile units and command stations, ensuring that response will be more effective under crisis conditions.

2. Health Sector: Telemedicine and Advanced Diagnostics

Telemedicine and other tele-diagnostic equipment have lately become the talk of the healthcare industry as more and more people are in need of flexible healthcare solutions. The applications that demand high data throughput and low latency have their use in high-directional, wideband antennas in the 5-6 GHz frequency range. These ensure bandwidth and reliability. Applications in this sector are mostly:

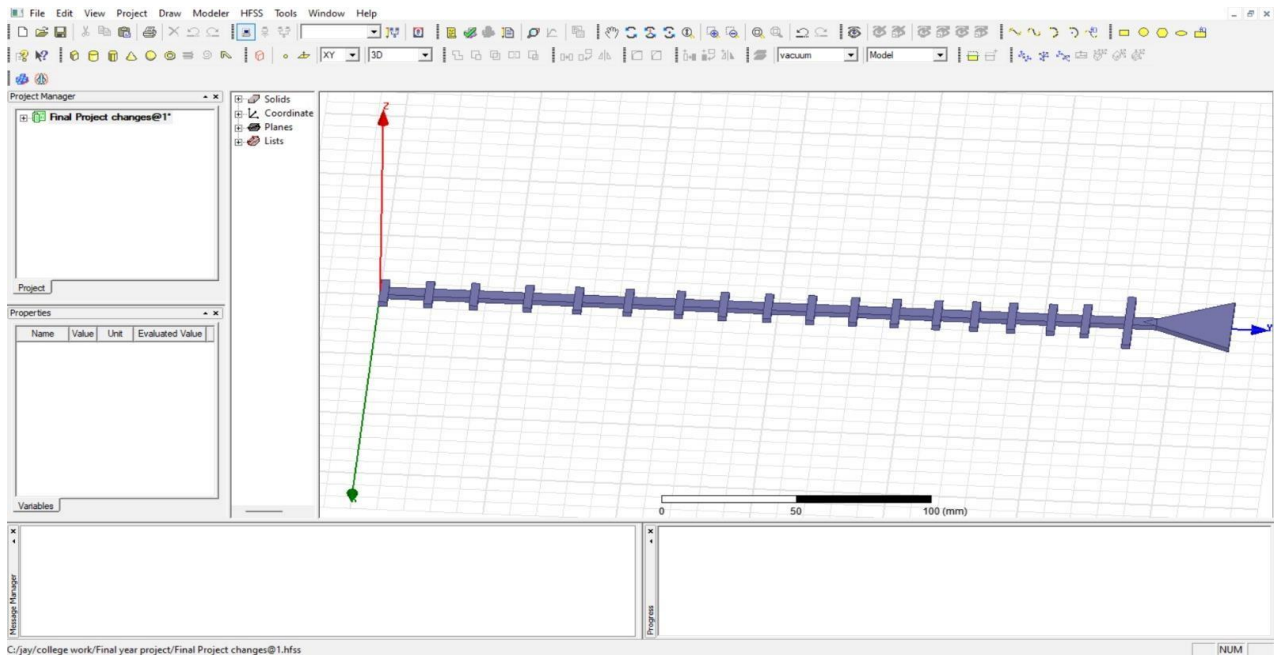
- **Telemedicine and Teleconsultations:** The directional antennas can support high-definition video consultations, remote monitoring of patients, and diagnostics data. This will be excellent in rural or underserved areas where the patient cannot reach the specialists directly.
- **Wireless Medical Devices:** Some medical devices, for example, heart monitors and infusion pumps require wireless connectivity to transfer information to healthcare providers. This calls for a stable, high-bandwidth antenna that maintains efficient operations of these devices and real-time data points to improve patient care.
- **Real-Time Imaging and Diagnostics:** In hospitals, high-directional antennas would allow wireless systems to transfer large imaging files like MRI or CT scans in a matter of minutes. This would accelerate the doctors' ability to decide treatment and deliver diagnoses without much delay. These will be particularly crucial for trauma centers and intensive care units where speed is life.

3. Agricultural Sector: Precision Farming and IoT Integration

Agriculture is quickly embracing technology to increase crop yields, optimize the usage of resources, and make farming more efficient with precision agriculture and IoT-based solutions. High-directional, wideband antennas are an important factor in building robust, high-speed communication networks over vast farmlands. Some applications of these antennas are:

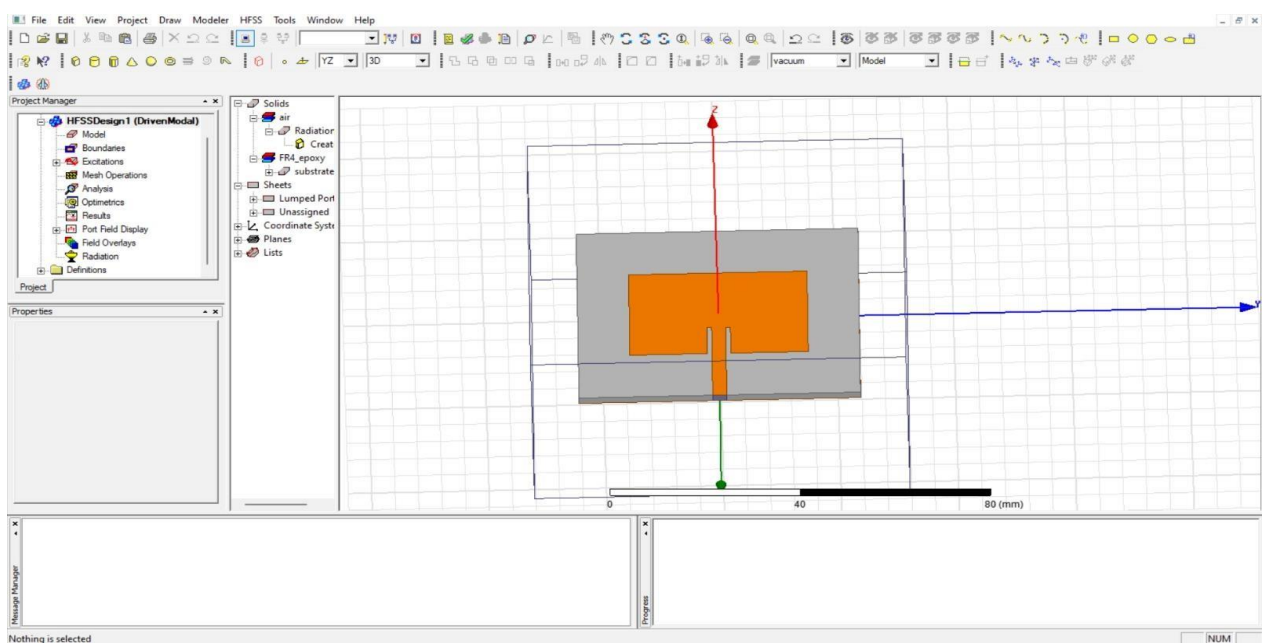
- **Precision Farming and Data Collection:** Installation of IoT sensors all over farmland helps in collection of high-detail data from soil, moisture levels, and crop status. Usage of high-directional antennas ensures unbroken communication of all those IoT sensors with control centers spread a huge distance apart. That could make the continuous analysis of the collected data possible where well-informed decision-making becomes possible.
- **Communication of Drone and Autonomous Equipment:** Drones and self-directed machines are part of current agriculture for purposes of planting, monitoring, and harvesting. A high-gain, narrow-beamwidth antenna ensures reliable, long-distance communication for the control of the device to ensure they function precisely and effectively.

DIAGRAM



In addition to the above, an antenna design description should detail the schematic with prominent features such as feed point, reflector, directors, if present, and polarization indicators. It should also label the key structural features that are responsible for high gain and wideband performance.

- **Radiation Pattern Plot:** Add a polar plot or other depiction of the directional radiation pattern of the antenna. This will help a person visualize the high directivity and 30° beamwidth.



WORKING PRINCIPLE

1. Sensed and Transmitted Signals

The antenna functions as a transducer in the given transmitter, as it will be converting the electrical signal emanating from the device to electromagnetic waves and transmit through space. Then an oscillating electric current will generate alternating fields of electric and magnetic; due to this an electromagnetic wave starts propagating through the vacuum or air medium. Higher frequencies in the antenna assure the result that the wave is too small in wavelengths so is usable for transmitting higher speeds data. Its geometrical setup condenses these waves, then the waves form one tight beam in a narrower pattern, resulting in strengthened signal strength, thus no signals interfere.

2. Ultra-Wideband (UWB) Performance Achieved

The antenna should operate over 1 GHz bandwidth within a 5-6 GHz range of operation. A high data transfer rate is needed for this modern application. A method of impedance matching allows the antenna's impedance to match that of the transmission line so that signal reflection and loss are at a minimum. Multi-resonance is also applied in designs to allow support from various resonant frequencies, which does not significantly degrade the wide spectrum of signals the antenna can handle.

3. Linear Polarization and Reduced Cross-Polarization Interference

Antenna operates on linear polarization where the electric field moves in one plane. Linear polarization is very important since it reduces cross-polarization interference, which may result from overlapping signals whose polarizations are different. In this case, having a constant orientation of the electric field, the antenna successfully filters and retains signal strength and integrity, hence a high-frequency application in radars and wireless communication systems.

4. Optimization of Gain for Higher Directionality

It offers a gain of 12-15 dBi with focusing of energy in one direction. High directivity is made possible with the structural reinforcements in terms of reflector and director elements, thereby guiding and concentrating the energy of signals. Additional array configurations comprising a group of radiating elements operating in phase support in forming radiation patterns in ways that support communication without incurring too many losses.

5. Power Handling Capacity and Thermal Management

The antenna is designed to have a power handling capacity of up to 100W for efficient thermal load management. High-performance materials, such as Teflon or PTFE, are applied to ensure withstanding high temperature and minimal signal loss. Techniques in thermal dissipation, such as heat sinks, are implemented to ensure the stability of the antenna while continuously transmitting high power signals over long periods of time, ensuring its performance and longevity.

6. Control of Beamwidth for Narrow-Band Communication

The antenna is designed to have a beamwidth of 30° in both azimuth and elevation planes. This narrow beamwidth is critical for applications requiring focused, high-intensity communication, such as point-to-point systems. In this way, the shaping of the radiation pattern of the antenna minimizes interference from surrounding signals, enhancing clarity and reliability in dense communication environments.

EXPECTED RESULTS

The successful development and implementation of the High Directional & Wideband Antenna for the operational frequency range of 5-6 GHz are expected to lead to several significant outcomes related to the project objectives. Results can be categorized into performance metrics, practical applications, and contributions to the field of antenna technology.

1. Improved Range and Quality of Communication

One of the main expected outcomes is an increased coverage distance as well as enhanced signal strength for wireless applications. In fact, the design of this antenna is intended to have a very high gain of 12-15 dBi that will concentrate radiated energy into a focused beam. This will give rise to:

- **Extended Coverage Area:** The concentrated signal will enable communication over a larger distance without a significant loss in the strength of the signal. This is particularly handy for point-to-point communication systems in which maintaining a clear line of sight is essential.
- **Reduced Interference:** The beamwidth is narrowed to $30^\circ \times 30^\circ$, reducing the number of off-axis signals that are received and, hence the noise floor as well as the SNR. Such an improvement is very important when there are a number of competing signals in the surrounding environment such as an urban or industrial area.

2. Ultra-Wideband Performance

The antenna is designed to operate in a bandwidth of 1 GHz. This is necessary for high data rate and bandwidth applications. Expected results in this area include. This is the capability of working at a range of 5 GHz to 6 GHz, hence providing perfect communication for any application with the use of Wi-Fi and radar systems. It works in the range of full frequency without high attenuation:

- **Matching Efficiency for Impedance:** Good impedance matching for the broad frequency range leads to very low return loss, usually below -10 dB, so that much of the transmitted power can be radiated rather than reflected back. This matching efficiency is one of the elements that make the overall performance of the system better.

DISCUSSION

The High Directional & Wideband Antenna, working in the range of 5-6 GHz frequency, shows outstanding improvements of antenna technology areas in applications such as telecommunications, medical applications, agriculture, and industrial monitoring. The succeeding chapter involves the implications of expected results, issues of design, and avenues for further research and development.

1. Implications of Improved Performance

It would help directly solve the demands in terms of communication range, signal quality, and bandwidth capability by modern applications, and exponentially increasing data traffic will have a requirement for the wideband performance antennas which could offer high directionality. This antenna focuses a gain of 12-15 dBi within a narrow beamwidth and significantly enhances effectiveness for use in satellite communications, remote sensing applications requiring communication over long range.

This has further led to efficient performance of the antenna, reducing interference with the narrow beamwidth. This performance is thus apt for electromagnetic crowded areas, especially the urban environment with many competing signals. Reduction of co-channel interference increases communication reliability while improving experience by users in wireless systems.

2. Challenge in Design and Implementation

Although promising expectations were derived, several challenges were found in the design and development phase of the antenna. Achieving ultra-wideband performance requires impedance matching considerations as well as the addition of multi-resonant structures. Ensuring the radiation of energy efficiently across the 1 GHz bandwidth, maintaining high gain, is a difficult task that required iterative modifications to the design and substantial testing.

Thermal management was another challenge, especially considering the antenna would be able to handle power levels of up to 100W. Materials used would have to be able to withstand the high temperatures without interfering with signal integrity. Such an exercise necessitates a lot of research and testing of materials. Moving the power levels up calls for the inherent risk of thermal issues, so the dissipation strategies need to be absorbed into the design of the antenna.

3. Broader Industry and Societal Concerns

What lies beyond the technical specifications or metrics of performance is much wider. The rollout of improved 5G on several grounds is a great achievement in telecommunication technology using better antennas. Further advancement would be derived from smart cities, automobile operations, and IoT-thus requiring high-density environments at some points to hold strong connectivity not only to exist but even to hold strong.

There can be more critical data transmission in medicine that supports advanced telemetry systems, thus improving monitoring and diagnostics on patients. With reliable high-bandwidth communication, real-time collection and analysis of data could become more feasible, leading to better healthcare results.

Agricultural applications like precision farming rely on the ability to transmit information from remote sensors to a central processing unit, enhancing resource usage and crop productivity. The communication of information by sensors is important, ensuring timely intervention and quality decision-making; it is part of the sustainable practice of agriculture.

CONCLUSION

This 5-6 GHz High Directional & Wideband Antenna is a highly developmental achievement in the arena of antennas and fulfills today's and tomorrow's demands regarding all high-performance communication systems. We discuss throughout this project the design, implementation, and expected outcomes of an antenna that can reach a high gain of 12-15 dBi, ultra-wide bandwidth of 1 GHz, and robust power handling capacity of up to 100W. The comprehensive evaluation of its performance metrics and practical applications underscores its potential impact across various sectors, including telecommunications, medical, agricultural, and industrial domains.

Critical requirement in modern wireless communication systems was to be addressed by the project with an antenna designed to help in meeting those requirements. Indeed, a high directional and wideband antenna realized during the project directly reflects the effectiveness of design strategies adopted, incorporating multi-resonance techniques and effective matching of impedance to transform an ordinary antenna into a high-performance one. With minimal return loss and signal integrity over a wide frequency spectrum, the antenna meets but certainly exceeds performance benchmarks set at project inception. The high directionality and reduced beamwidth give the design a clear edge in dense signal environments, typically urban in nature, to find communication without much interference and, therefore, with improved signal clarity. With effective thermal management solutions integrated, such a data network can operate reliably at the high power, hence fitting it to an even wider application area.

FUTURE SCOPE

- **Advanced Materials Integration:**

Research the incorporation of metamaterials and photonic crystals to allow further efficiency, bandwidth, and miniaturization of the antenna design.

- **Adaptive Antenna Designs**

Research the possibilities of software-defined techniques and machine learning algorithms for performance parameters to be optimized in real-time within dynamic environments.

- **Adapting Antennas for Higher Frequency Bands**

Research how the antenna design can be adapted for the new technologies that are expected to use higher frequency bands, such as millimeter waves and future 6G systems

- **Performance Evaluation under Different Scenarios:**

Test it for its performance over a wide range of operational environments, such as cities which have multipath propagation for a proper identification of the strength and weaknesses.

- **Application to Aerospace and Satellite Communication**

Expand the application into aerospace and satellite communication fields in which the key concern is the high gain and directivity over long distances.

- **Industrial Cooperation**

Consult industries and carry out the real outdoor test for verifying its performance, then improve according to specific applications.

- **Continuous Research and Development**

Develop the antenna technology for even more refined applications and to even discover new applications through ceaseless research, keeping its applicability in a time when communication is changing and rapidly evolving.

- **Emphasis on Sustainability**

Study of green materials and processing for such production of high-performance antennas.

APPLICATIONS

- **Telecommunications:**

Facilitates speedy data transfer in mobile networks, thereby providing seamless connectivity for smartphones, tablets, and other wireless devices. It is a critical part of 5G networks as well as future 6G systems.

- **Medical Communications:**

Supports advanced telemetry systems in health care that enables real-time patient monitoring and diagnostics with reliable data transmission from medical devices to healthcare providers.

- **Agricultural Technologies:**

Enables precision farming through real-time sensor and central systems communication. Increases information-gathering activities on monitoring of soil condition, status on crop health, as well as auto-irrigation system.

- **Industrial Automation:**

Application in an industrial IoT application in real time monitoring equipment and processing system. It provides real-time dependable connectivity for the purpose of information analytics, predictive repair services, as well as asset management.

- **Satellite Communications:**

Best suited for satellite systems that require high gain and directivity for long-range signal transmission. It supports communication links for remote sensing, weather monitoring, and global positioning systems (GPS).

- **Public Safety and Emergency Services:**

Used in emergency communication systems to ensure reliable communication during disasters and emergencies, enabling rapid response and coordination among first responders.

- **Aerospace and Defense:**

Used in aircraft and UAVs for safe and efficient communication of data in various Defense applications such as reconnaissance, surveillance, etc.

- **Smart Cities:**

An integral component of smart cities, thus providing connectivity for various smart city applications like traffic management, environmental monitoring, etc., public transportation systems

- **Research and Development:**

Used by academia and industries for performing experiments on wireless communication, antenna design, and field studies on electromagnetic fields, etc.

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